

CetaSignal: Constraint-Based Acoustic Classification of Cetacean Coordination Signals

Research Prototype Brief for Scientific Discussion

CetaSignal is a rule-based acoustic analysis framework designed to classify whale vocalizations according to coordination behaviors rather than species-specific call labels. The premise is that whale communication evolved under the physical constraints of the ocean—dense water, low visibility, long-distance acoustic propagation, and three-dimensional spatial navigation. These constraints shaped vocal signals optimized for behavioral coordination such as maintaining pod cohesion, synchronizing dives, coordinating foraging, long-distance navigation, and broadcast display signals.

Constraint Framework: The ocean environment imposes several communication constraints that favor acoustic signaling:

- Water transmits sound efficiently compared to light.
- Visibility is limited, often <50 meters.
- The ocean is a 3-dimensional environment requiring depth coordination.
- Low frequency sounds can propagate across basin scales via the SOFAR channel.

These constraints form the theoretical basis of the CetaSignal classifier.

Dataset Overview: The current research prototype evaluates approximately 5000 acoustic specimens spanning 19 cetacean species across multiple ocean basins. Source datasets include publicly available hydrophone archives and marine mammal acoustic libraries. Each specimen includes an associated behavioral observation recorded independently from the acoustic signal. During evaluation, the classifier was provided only acoustic features and generated behavior predictions without access to the labeled outcome.

Acoustic Feature Extraction

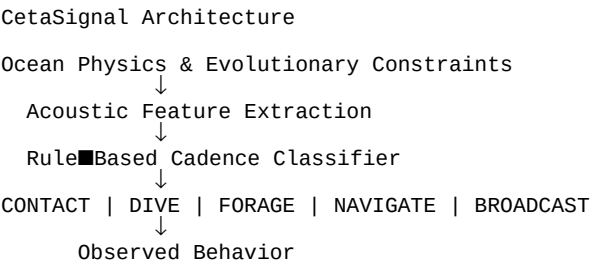
Feature	Measurement Method
Peak Frequency	Highest power frequency detected in the signal window via FFT spectrogram
Frequency Contour	Slope of peak frequency across time (rising, descending, flat, complex)
Call Duration	Signal end time minus signal start time
Inter-Pulse Interval (IPI)	Time between pulses in pulsed calls
Repetition Rate	Calls per unit time
Urgency Index	$(\text{Pulse density} \times \text{repetition rate}) / \text{signal duration}$

Behavioral Coordination Classes

Behavior Class	Description
CONTACT	Signals maintaining pod cohesion

DIVE	Short descending signals coordinating descent
FORAGE	High urgency pulses during prey detection
NAVIGATE	Low frequency long-range migration coordination
BROADCAST	Long complex signals often associated with reproductive display

System Architecture



Classification Model: The CadenceClassifier uses transparent decision rules derived from biological and physical constraints. Example rule: IF peak frequency < 200 Hz AND repetition interval > 2 seconds → NAVIGATE signal Unlike deep neural networks, the rule-based model preserves interpretability, allowing researchers to trace each classification to specific acoustic features.

Future Research

Future Research Directions:

- Integration with tag-based whale movement data (DTAG datasets).
- Playback experiments to test predicted behavioral responses.
- Analysis of juvenile vocal development and learning patterns.
- Real-time hydrophone monitoring for behavioral state detection.